

1 The Distance Route: Single-Stage Spectral Recovery

This section establishes the exact recovery of tree splits using the sign pattern of the leading eigenvector of the double-centered distance matrix. Rather than relying on multi-stage refinement algorithms or demanding full knowledge of the eigenvector’s shape, we pursue a purely spectral approach. By importing the structural Schur complement theory of Stone and Griffing [6] and the leave-one-out entrywise perturbation framework of Abbe et al. [1], the recovery guarantee emerges organically from the native geometry of the distance matrix.

1.1 Population Geometry and the Spectral Target

Let T be a weighted tree with m leaves, and let $\mathcal{D}$ be the additive path-distance matrix between the leaves. We apply double centering with the orthogonal projector $H = I - \frac{1}{m}\mathbf{1}\mathbf{1}^\top$ to form the distance operator:

$$\mathcal{B} := HDH \tag{1}$$

The recovery algorithm classifies leaves by the sign pattern of $u^{(1)}$, the unit eigenvector of $\mathcal{B}$ belonging to its largest-magnitude eigenvalue.

Griffing [3] showed that $-\frac{1}{2}\mathcal{B}$ is exactly the Moore-Penrose pseudoinverse of the tree Laplacian Schur-complemented onto the leaves. Consequently, $u^{(1)}$ is mathematically identical to the Fiedler vector of this Schur complement. By the structural theorems of Stone and Griffing [6], this vector unconditionally partitions the leaves into a valid split of the tree.

Lemma 1 (Target Geometry [6]). *The sign pattern of $u^{(1)}$ corresponds to a bipartition of the leaves cut by a single edge in T . Furthermore, $u^{(1)}$ has no zero coordinate, yielding a strictly positive margin $\delta_* := \min_i |u_i^{(1)}| > 0$.*

Because we do not assume the tree is perfectly symmetric or balanced, $u^{(1)}$ is not strictly piecewise constant. We treat its geometric “bumpiness” simply as a parameter of the margin δ_* . Furthermore, we assume $u^{(1)}$ is delocalized such that $\delta_* \geq c_\delta/\sqrt{m}$ and $\|u^{(1)}\|_\infty \leq C_\delta/\sqrt{m}$.

1.2 Observation Model and Error Decomposition

We observe each pairwise distance independently with probability p . The inverse-probability weighted empirical matrix is $\hat{\mathcal{D}} = p^{-1}\mathcal{D} \circ \Omega$, where $\Omega_{ij} \sim \text{Bernoulli}(p)$. The empirical distance operator is $\hat{\mathcal{B}} = H\hat{\mathcal{D}}H$, with leading eigenvector $\hat{u}^{(1)}$.

The error takes the form $\hat{\mathcal{B}} = \mathcal{B} + E_{\mathcal{B}}$, where $E_{\mathcal{B}} = H(\hat{\mathcal{D}} - \mathcal{D})H$ is the zero-mean sampling noise. Let $E_{\mathcal{D}} = \hat{\mathcal{D}} - \mathcal{D}$ be the uncentered noise matrix, which has independent mean-zero entries bounded by $d_{\max}/p$, and variance bounded by $d_{\max}^2/p$.

1.3 Full Entrywise Proof

Theorem 1 (Single-Stage Exact Recovery). *Assume the spectral gap $\Delta\lambda = |\lambda_1(\mathcal{B})| - |\lambda_2(\mathcal{B})|$ is strictly positive. If the sampling rate satisfies*

$$p = \Theta\left(\frac{d_{\max}^2}{\Delta\lambda^2 \cdot \delta_*^2} \log m\right) \tag{2}$$

then with high probability, $\text{sign}(\hat{u}^{(1)}) = \text{sign}(u^{(1)})$, meaning the raw spectral signs exactly recover the split. For a well-behaved tree where $\Delta\lambda \asymp m$ and $\delta_ \asymp 1/\sqrt{m}$, this yields the standard $p = \Theta(\frac{\log m}{m})$ rate.*

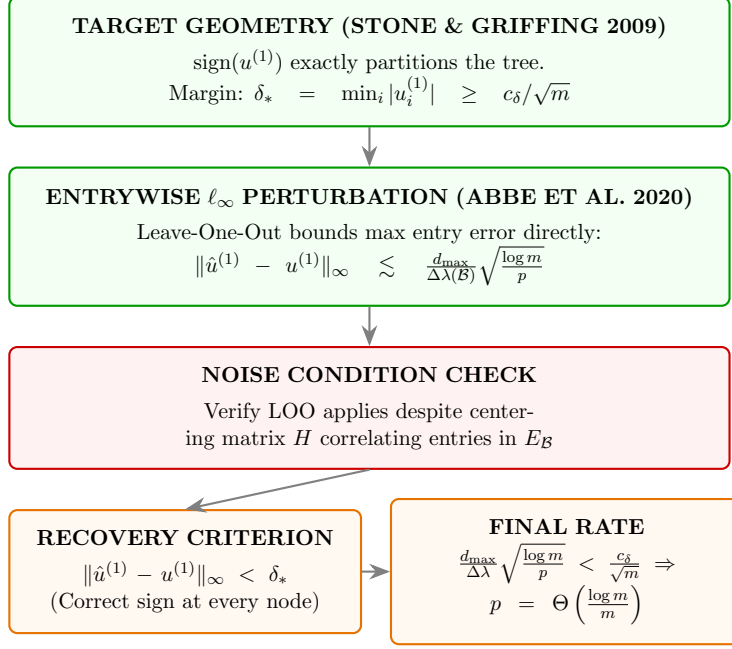


Figure 1: The Leave-One-Out (LOO) proof route for single-stage exact recovery.

Proof. The proof proceeds by bounding the entrywise ℓ_∞ error of the eigenvector directly, bypassing the weak ℓ_2 dimension factor. We deploy the Leave-One-Out (LOO) perturbation framework. For clarity, we restate the relevant consequence of the master theorem from Abbe et al. [1]:

Theorem 2 (Abbe et al. 2020, Theorem 2.1 (Consequence for Symmetric Matrices)). *Let M be a symmetric matrix with eigendecomposition $M = U\Lambda U^\top$, and let $\hat{M} = M + E$ where E is a symmetric noise matrix with independent mean-zero entries (up to symmetry). Let $\hat{U}$ denote the empirical top r eigenvectors. Provided the spectral gap $\Delta\lambda$ is sufficiently large, there exists an orthogonal alignment matrix W such that:*

$$\|\hat{U}W - U\|_\infty \lesssim \frac{\|EU\|_\infty}{\Delta\lambda} + \frac{\|E\|_2 \|E\|_{\text{row-max}}}{\Delta\lambda^2} \|U\|_\infty \quad (3)$$

In our context, the target signal is the rank-1 leading subspace ($r = 1$), so U is the single vector $u^{(1)}$, W is a scalar sign flip (which we resolve by assuming aligned signs), M is the population distance operator $\mathcal{B}$, and E is the centered noise matrix $E_{\mathcal{B}}$. While $\mathcal{B}$ is full rank, the LOO expansion strictly isolates the leading eigenspace gap $\Delta\lambda$, yielding the analogous bound:

$$\|\hat{u}^{(1)} - u^{(1)}\|_\infty \lesssim \frac{\|E_{\mathcal{B}}u^{(1)}\|_\infty}{\Delta\lambda} + \frac{\|E_{\mathcal{B}}\|_2 \|E_{\mathcal{B}}\|_{\text{row-max}}}{\Delta\lambda^2} \|u^{(1)}\|_\infty \quad (4)$$

We now rigorously bound each of the relevant norms for the centered noise matrix $E_{\mathcal{B}} = HE_{\mathcal{D}}H$.

1. Bounding the first-order term $\|E_{\mathcal{B}}u^{(1)}\|_\infty$:

Since the population vector is centered ($Hu^{(1)} = u^{(1)}$), we have $E_{\mathcal{B}}u^{(1)} = HE_{\mathcal{D}}u^{(1)}$. The i -th entry of $E_{\mathcal{D}}u^{(1)}$ is:

$$(E_{\mathcal{D}}u^{(1)})_i = \sum_{j=1}^m (E_{\mathcal{D}})_{ij} u_j^{(1)} \quad (5)$$

Because the entries $(E_{\mathcal{D}})_{ij}$ are independent with mean zero, and $u^{(1)}$ is a fixed vector, this is a sum of independent random variables. The variance of the sum is bounded by $\sum_j \frac{d_{\max}^2}{p} (u_j^{(1)})^2 = \frac{d_{\max}^2}{p}$.

The maximum absolute value of a single summand is bounded by $\frac{d_{\max}}{p} \|u^{(1)}\|_\infty$. By Bernstein's inequality, with high probability:

$$\|E_{\mathcal{D}} u^{(1)}\|_\infty \lesssim d_{\max} \sqrt{\frac{\log m}{p}} \quad (6)$$

The centering matrix H only subtracts the mean $\frac{1}{m} \mathbf{1}^\top E_{\mathcal{D}} u^{(1)}$, which is an average of these concentrated sums and is therefore of even smaller order. Thus, $\|E_{\mathcal{B}} u^{(1)}\|_\infty \lesssim d_{\max} \sqrt{\frac{\log m}{p}}$.

2. Bounding the operator and row norms ($\|E_{\mathcal{B}}\|_2$ and $\|E_{\mathcal{B}}\|_{\text{row-max}}$):

By standard Matrix Bernstein inequalities [?], the operator norm of the independent noise matrix concentrates as $\|E_{\mathcal{D}}\|_2 \lesssim d_{\max} \sqrt{\frac{m}{p}}$. Because H is a projector, $\|E_{\mathcal{B}}\|_2 \leq \|E_{\mathcal{D}}\|_2 \lesssim d_{\max} \sqrt{\frac{m}{p}}$. Similarly, the row norm $\|e_i^\top E_{\mathcal{B}}\|_2$ is dominated by the Euclidean norm of a row of $E_{\mathcal{D}}$, which is the square root of m independent variables each with variance $d_{\max}^2/p$. By scalar concentration, $\|E_{\mathcal{B}}\|_{\text{row-max}} \lesssim d_{\max} \sqrt{\frac{m}{p}}$.

3. Synthesis and Recovery Criterion:

Substitute these probabilistic bounds into the LOO master equation:

$$\|\hat{u}^{(1)} - u^{(1)}\|_\infty \lesssim \frac{d_{\max}}{\Delta\lambda} \sqrt{\frac{\log m}{p}} + \frac{d_{\max}^2 m}{p \Delta\lambda^2} \|u^{(1)}\|_\infty \quad (7)$$

For exact sign recovery, we require this total deviation to be strictly less than the margin δ_* . Assuming standard delocalization where $\|u^{(1)}\|_\infty \asymp 1/\sqrt{m}$ and $\delta_* \asymp 1/\sqrt{m}$, the two terms require:

$$\text{Term 1: } \frac{d_{\max}}{\Delta\lambda} \sqrt{\frac{\log m}{p}} < \frac{c_\delta}{\sqrt{m}} \implies p \gtrsim \frac{d_{\max}^2 m \log m}{\Delta\lambda^2} \quad (8)$$

$$\text{Term 2 (H.O.T.): } \frac{d_{\max}^2 m}{p \Delta\lambda^2 \sqrt{m}} < \frac{c_\delta}{\sqrt{m}} \implies p \gtrsim \frac{d_{\max}^2 m}{\Delta\lambda^2} \quad (9)$$

Since $\log m > 1$, the first-order term dominates the sample complexity. Thus, $p = \Theta\left(\frac{d_{\max}^2 m \log m}{\Delta\lambda^2}\right)$ is sufficient for exact sign recovery. $\square$

1.4 Open Question: How Do $\Delta\lambda$ and δ_* Depend on the Tree?

The rate above is written in terms of two numbers, and both are fixed as soon as the tree is fixed — no sampling enters either one:

- the gap $\Delta\lambda = |\lambda_1(\mathcal{B})| - |\lambda_2(\mathcal{B})|$, which says how far the leading direction of $\mathcal{B}$ stands out from the next one;
- the smallest entry $\delta_* = \min_i |u_i^{(1)}|$, which says how close the closest leaf comes to sitting on the fence between the two sides.

We put $\Delta\lambda \asymp m$ and $\delta_* \asymp 1/\sqrt{m}$ into the general rate to get $p \gtrsim \log m/m$. What is open is how to read those two numbers off a tree:

Given a weighted tree T with m leaves, how small can $\Delta\lambda$ and δ_* be, in terms of the shape of T and the lengths of its branches?

The two travel together. The vector $u^{(1)}$ is the direction that maximises the Rayleigh quotient of $\mathcal{B}$, so a tree that leaves little room between the first and second directions is also a tree whose leading vector comes close to zero somewhere. A useful reference point: if $u^{(1)}$ were exactly two-valued, one value on each side of a split that puts a fraction $\alpha \leq 1/2$ of the leaves on the smaller side, then centering and unit length fix its entries, and the smaller of the two is

$$\delta_* = \frac{1}{\sqrt{m}} \sqrt{\frac{\alpha}{1-\alpha}}. \quad (10)$$

So the margin is set by how lopsided the split is, and α is something the tree model decides. Table 1 collects what each kind of tree does to the two numbers.

Tree	$\delta_* \approx$	$\Delta\lambda \approx$	Note
Balanced binary	$1/\sqrt{m}$	order m	Symmetry fixes the vector: it is close to two-valued with $\alpha = 1/2$, so every entry has about the same size.
Kingman coalescent	$\sqrt{\alpha/(1-\alpha)}/\sqrt{m}$	a constant fraction of $ \lambda_1 $	Theory: the topology is Yule, whose root split is uniform in size [5, 2], so the smaller side α is uniform on $\{1/m, \dots, 1/2\}$; it is typically of order one and δ_* of order $1/\sqrt{m}$, with a tail down to $1/m$ when the split peels off a single leaf. The last coalescence lasts $\text{Exp}(1)$ against a tree height of mean $2(1 - 1/m)$ [4, 7], so the deepest split stands clear of the next one by a fixed proportion. Both are borne out in Figures 4 and 5.
Pure birth (Yule branch lengths)	$0.024/\sqrt{m}$	$0.38 \lambda_1 $	Same topology law as the row above, so the same α ; what changes is where the branch lengths sit. Medians over 200 trees at $m = 1000$, against $0.36/\sqrt{m}$ and $0.60 \lambda_1 $ for the coalescent; see Figures 4 and 5.
Caterpillar, very uneven branch lengths (extreme)	well below $1/\sqrt{m}$	well below m	The vector puts most of its weight on a handful of leaves and the rest of the entries are tiny, which is $\alpha \rightarrow 1/m$ in (10). How far below is the open part.

Table 1: What each kind of tree does to the two numbers the rate is written in.

So one half of the assumption in Section 1.1 is comfortable and the other half is where the work

is. The bound $\|u^{(1)}\|_\infty \leq C_\delta/\sqrt{m}$ holds on both random models with a small constant. The margin $\delta_* \geq c_\delta/\sqrt{m}$ is the half that a tree can push down, and since p scales as $1/\delta_*^2$, a margin ten times smaller asks for a hundred times more of the matrix. The general form $p \gtrsim d_{\max}^2 \log m / (\Delta \lambda^2 \delta_*^2)$ covers all three rows; what each row needs is the right pair of numbers to put into it.

A The two operators on simulated trees

The quantities the recovery rate is stated in, $\Delta \lambda$ and δ_* , are properties of the population operator, so they can be read off simulated trees exactly. The figures below do that on Jukes–Cantor sequences of length $L = 10^4$ over trees with $m = 1000$ leaves: one tree for the anatomy of a single split (Figures 2 and 3), and pools of 200 Kingman and 200 birth–death trees for the distributions (Figures 4 and 5). Throughout, $\mathcal{B}$ is compared against the graph Laplacian $L(S) = \text{Deg}(S) - S$ of the similarity matrix $S = e^{-\mathcal{D}}$, whose Fiedler vector is the alternative classifier.

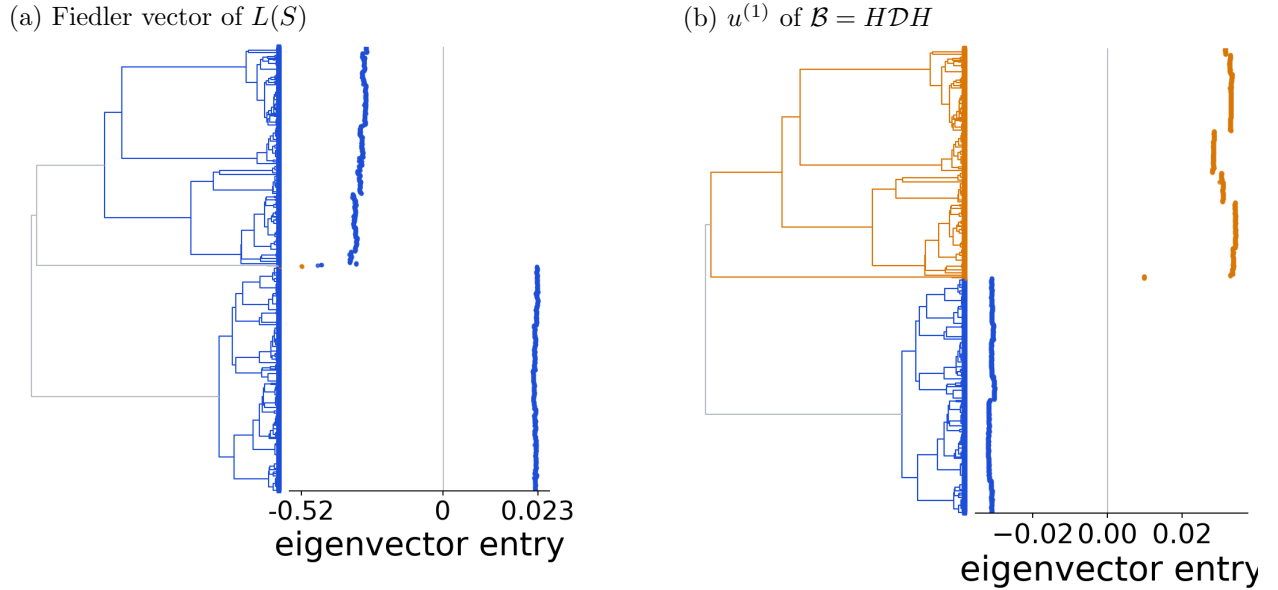


Figure 2: One Kingman tree, drawn twice, each copy cut by its own vector: blue and orange are the two sides of that cut, grey a clade straddling it, and the scatter holds the vector entries in the same leaf order. The tree shown is the one in the pool where the two cuts differ most: $u^{(1)}$ splits it 505/495, the Fiedler vector 998/2.

The two vectors are read off two different matrices built from the same alignment, shown next in the same leaf order.

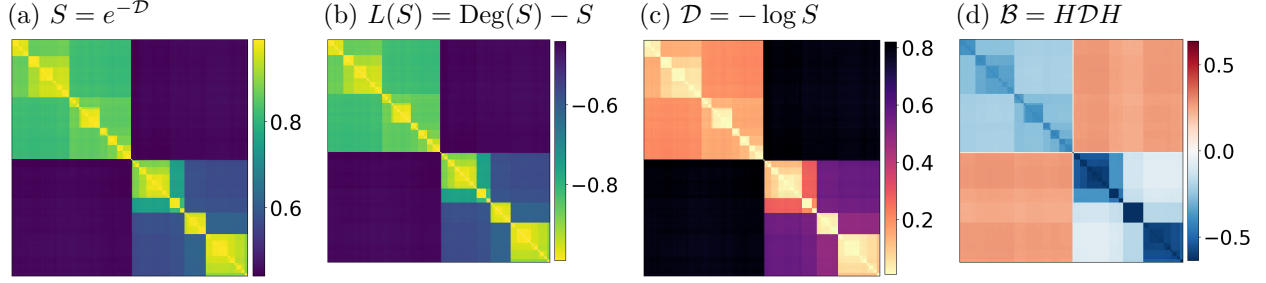


Figure 3: The four matrices on the tree of Figure 2, leaves ordered so that a clade is a contiguous block. Each panel carries its own colour scale: the diagonal of $L(S)$ is the degree, three orders above its off-diagonal, and is masked in grey; $\mathcal{B}$ is signed and shown diverging about zero.

Across the pools, the quantity that governs the rate is the separation between the first two eigenvalues.

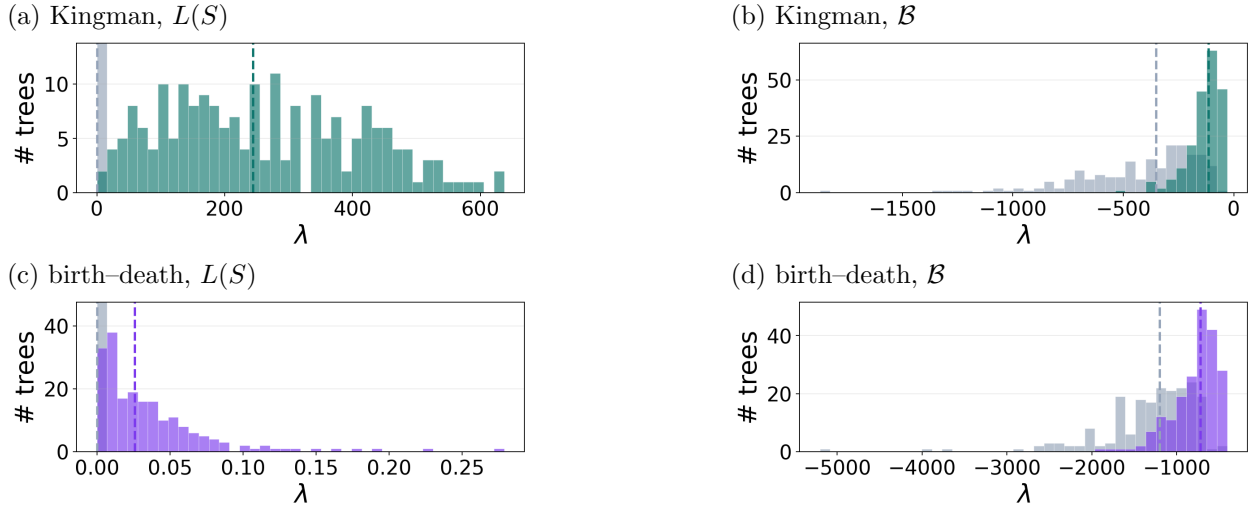


Figure 4: Distribution over 200 trees per model of the two leading eigenvalues of each operator: grey λ_1 , coloured λ_2 , dashed lines at the medians. For $L(S)$ the eigenvalues are ascending and $\lambda_1 \equiv 0$, so its bar is clipped; for $\mathcal{B}$ they are the two largest in magnitude, both negative. The relative gap $(|\lambda_1| - |\lambda_2|)/|\lambda_1|$ of $\mathcal{B}$ has median 0.60 on Kingman trees and 0.38 on birth-death trees, against 0.20 and 0.25 for the corresponding Fiedler gap $(\lambda_3 - \lambda_2)/\lambda_3$.

The margin δ_* is a property of the entries themselves, pooled below over the same trees.

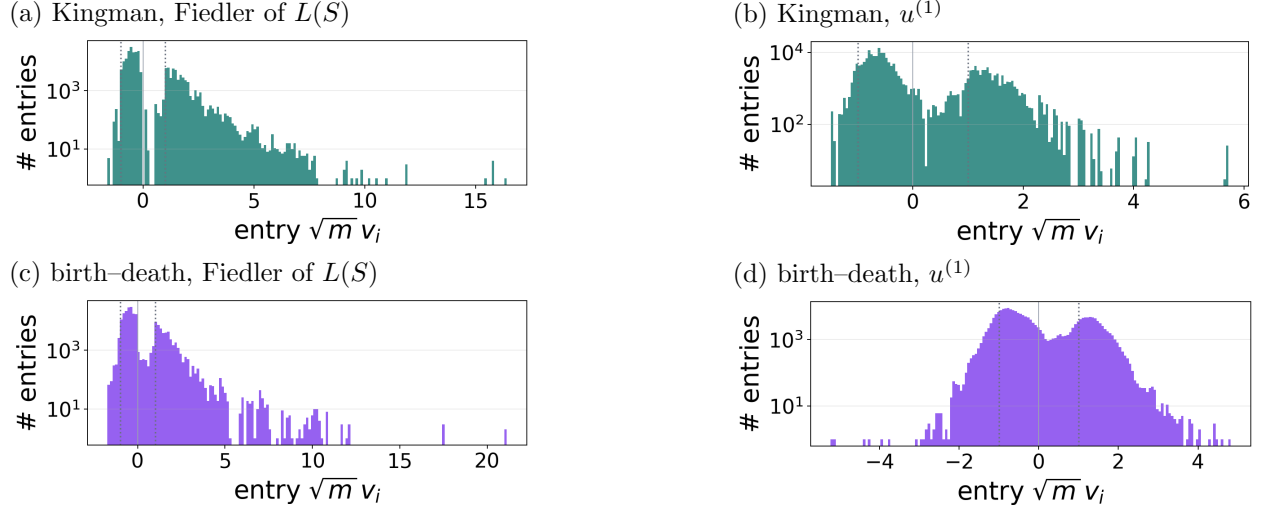


Figure 5: Entries of the classifying vector, pooled over the same 200 trees per model, each vector normalised to $\|v\| = 1$, scaled by $\sqrt{m}$ and oriented with the minority side positive; counts are logarithmic and the dotted lines mark ± 1 , the value a perfectly delocalised vector takes. The entries of $u^{(1)}$ cluster at ± 1 with a largest entry of 5.7 against $\sqrt{m} = 31.6$, which is the delocalisation $\|u^{(1)}\|_\infty \asymp 1/\sqrt{m}$ assumed in Section 1.1; the Fiedler vector reaches 16.4 on Kingman trees and 21.1 on birth-death trees.

References

- [1] Emmanuel Abbe, Jianqing Fan, Kaizheng Wang, and Yiqiao Zhong. Entrywise eigenvector analysis of random matrices with low expected rank. *Annals of Statistics*, 48(3):1452–1474, 2020.
- [2] David J Aldous. Stochastic models and descriptive statistics for phylogenetic trees, from Yule to today. *Statistical Science*, 16(1):23–34, 2001.
- [3] Alexander Griffing. *Connections Between Numerical Taxonomy and Phylogenetics*. PhD thesis, North Carolina State University, 2012.
- [4] John Frank Charles Kingman. The coalescent. *Stochastic Processes and their Applications*, 13(3):235–248, 1982.
- [5] Mike Steel and Andy McKenzie. Properties of phylogenetic trees generated by yule-type speciation models. *Mathematical Biosciences*, 170(1):91–112, 2001.
- [6] Eric A Stone and Alexander R Griffing. On the fiedler vectors of graphs that arise from trees by schur complementation of the laplacian. *Linear Algebra and its Applications*, 431(10):1869–1880, 2009.
- [7] John Wakeley. *Coalescent Theory: An Introduction*. Roberts and Company, 2009.